Unit 3 (Part 4) - Sets

Representation Of Number Sets

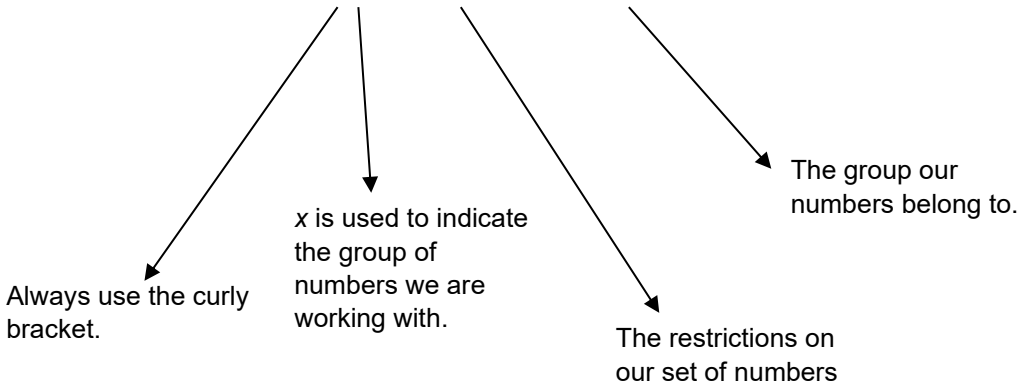
There are different ways that a set of numbers can be represented. Following are the most commonly used methods. You must know them and how to change from one form to another.

⇒ Listing the elements: Here the actual numbers are all written down in set brackets { }

Example 1: All the whole numbers between 4 and 11. {5;6;7;8;9;10}

Set Builder Notation: Here a set is built up using mathematical notation.

$$\{x : 4 < x < 11 ; x \in N_0\}$$



Write the following in set builder notation:

Example 1: All the whole numbers between 4 and 11.

Example 2: All the Integers less than or equal to 4.

Example 3: A class of learners must contain at least 10 learners and a maximum of 30 learners.

Interval Notation: Interval notation is the shortest and most widely used notation. There are however some things to remember.

This notation may **only** be used when dealing with **real numbers**.

The smaller value is always written on the left.

A square bracket, [or], is used when the number is to be included in the set.

A round bracket, (or), is used when the number is not included in the set.

∞ and $-\infty$ are always used in conjunction with a round bracket, the logic being, that you can never reach infinity, therefore you cannot include it.

If you have a split set, then the sets need to be united with the union symbol, $\cup$.

Write the following in interval notation:

Example 1: All real numbers between -11 and 7.

Example 2: All real numbers greater than or equal to -6.

Example 3: Real numbers less than or equal to -4 and greater than 6.

Numberline Notation: This is also known as a graphical representation.

Number lines are drawn with the numbers increasing from left to right.

The scale has to be appropriately chosen and consistent throughout the number line.

An arrow is drawn at the end of the line to show the direction in which the numbers are increasing.

For N, N_0 and Z a solid dot, •, is placed above each number on the number line.

For Real numbers, dots are placed above the end points and they are joined by a solid line.

If the end number is included, a solid dot, •, is placed above the number.

If the end number is not included, an open dot, ◦, is placed above the number.

<u>Complete the following showing answers on a number line.</u>

Example 1: Natural numbers from 6 to 10

Example 2: The set of Integers less than 4

Example 3: The set of real numbers less than 2 and greater than or equal to 4.

Summary

Complete the table below

Notation	Included	Not Included	Number Group
Set Builder		< or >	
Interval	[or]		
Number line			Solid Line for R Dots on each for N, No and Z

Setbuilder Notation

Setbuilder notation is a useful way of representing sets of real numbers.

Examples:

Represent the following sets on a number line:

a. $\{x: x \geq 4; x \in \mathbb{Z}\}$

b. $\{x: x < 4; x \in \mathbb{Z}\}$

c. $\{x: x < 5; x \in \mathbb{N}\}$

d. $\{x: -4 < x \leq 3; x \in \mathbb{Z}\}$

e. $\{x: x \geq 2; x \in \mathbb{R}\}$

f. $\{x: x < 8; x \in \mathbb{R}\}$

g. $\{x: -5 \leq x < 4; x \in \mathbb{R}\}$

Interval Notation

Interval notation is another way of representing real numbers on the number line.

Examples:

Represent the following on a number line

a. $a \in (-1; 3\,]$

b. $b \in [-5; 6)$

c. $c \in [3; \infty)$

d. $d \in (-\infty; 4)$

Remember:

Infinity is not a number so it CANNOT be included [ ] in Interval notation.

When using Interval Notation, the smaller of the two numbers is ALWAYS on the left of the bigger number.

Homework: Ex. 1 (**D – G**) pg. 7 (Mind Action Series new edition, 2022)